

3.2.2 B: Complex numbers

	Content
B1	Solve any quadratic equation with real coefficients; solve cubic or quartic equations with real coefficients (given sufficient information to deduce at least one root for cubics or at least one complex root or quadratic factor for quartics).

	Content
B2	Add, subtract, multiply and divide complex numbers in the form $x + iy$ with x and y real; understand and use the terms 'real part' and 'imaginary part'.

	Knowledge/skill
B4	Use and interpret Argand diagrams.
	Content
B5	Convert between the Cartesian form and the modulus-argument form of a complex number (knowledge of radians is assumed).
	Content
B6	Multiply and divide complex numbers in modulus-argument form (knowledge of radians and compound angle formulae is assumed).
	Content
B7	Construct and interpret simple loci in the Argand diagram such as $ z - a > r$ and $\arg(z - a) = \theta$ (knowledge of radians is assumed).
	Knowledge/skill
B8	Understand de Moivre's theorem and use it to find multiple angle formulae and sums of series.
	Content
B9	Know and use the definition $e^{i\theta} = \cos\theta + i\sin\theta$ and the form $z = re^{i\theta}$
	Content
B10	Find the n distinct n th roots of $re^{i\theta}$ for $r \neq 0$ and know that they form the vertices of a regular n -gon in the Argand diagram.
	Content
B11	Use complex roots of unity to solve geometric problems.